

Questions 31–40

Complete the notes below. Write **ONE WORD ONLY** for each answer.

New engineering designs: Scottish case study

Project to renovate and link two canals in Scotland

Historical background

- **1790:** opening of the Forth and Clyde Canal (Scotland's **31** _____ canal)
- **1822:** opening of the Union Canal
- The two canals allowed boats to cross the **32** _____ part of Scotland
- Locks (steps) connected the two canals near Falkirk
- **1842:** competition from a new **33** _____
- **1930s:** the locks were demolished

Millennium project

- Aim: to improve Scotland's **34** _____
- The long-term goal – to strengthen the Scottish **35** _____
- Involved restoration of the two canals
- Problem: one canal is 35m lower than the other

The Falkirk Wheel (a mechanical lift)

- Completed in 2002
- Two large tanks are attached to the ends of an arm
- The tanks are designed to remain in a **36** _____ position
- A boat sails into each tank
- The weight of the tanks with their contents is always **37** _____
- Little energy is needed to rotate the arm
- Visual quality – resembles a **38** _____

Manufacturing

- All the parts were constructed off-site
- 1,200 tonnes of **39** _____ were used
- Joints were bolted and welded to reduce the effects of **40** _____

- 31. **oldest**
- 32. **central**
- 33. **railway**
- 34. **transport**
- 35. **economy**
- 36. **horizontal**
- 37. **equal**
- 38. **sculpture**
- 39. **steel**
- 40. **stress**